

PHYS 708 – Assignment 1 (Due date: 19 September 2025)

Some useful relation and constants:

Total energy of a particle, $E = \gamma m_0$, where γ is the Lorentz factor, m_0 is the rest mass energy.

$E^2 = p^2 + m_0^2$, where p is the total momentum of the particle in units of energy/ c , c the velocity of light.

Neutral pion rest mass energy = 135 MeV.

Proton rest mass energy = 938 MeV.

Electron rest mass energy = 0.5 MeV.

Energy of a cosmic microwave background photon = 7×10^{-4} eV.

Problem 1

(a) Obtain an expression for the threshold kinetic energy of an incident proton to produce a neutral pion in a proton-proton interaction: $p + p \rightarrow p + p + \pi^0$.

Hint: Assume the target proton to be at rest, and use invariant property of the mass. Then, equate the invariant mass obtained for the two sides of the equation by considering lab frame before the collision and center of mass frame after the collision. The threshold kinetic energy will be in the form:

$$E_k = \frac{m_{\pi^0}}{2m_p}(m_{\pi^0} + 4m_p),$$

where m_{π^0} and m_p are the pion and proton rest mass energies.

(b) Find the value of E_k for the interaction.

Problem 2

(a) Show that the threshold Lorentz factor of an incident proton to produce an e^-e^+ pair in a proton-photon interaction ($p + \text{photon} \rightarrow p + e^- + e^+$) is given by $\gamma_{\text{th}} \approx m_e/\epsilon$, where m_e is the rest mass energy of electron and ϵ is the target photon energy. Consider the collision to be head-on.

(b) Calculate the threshold total energy of the incident proton for an interaction on a cosmic microwave background photon.

Hint: Similar to that of Question 1.

Problem 3

(a) Show that the threshold Lorentz factor of an incident proton to produce a neutral pion in a proton-photon interaction ($p + \text{photon} \rightarrow p + \pi^0$) is given by:

$$\gamma_{\text{th}} \approx \frac{m_{\pi^0}}{2\epsilon} \left(1 + \frac{m_{\pi^0}}{2m_p} \right),$$

where m_{π^0} is the rest mass energy of the neutral pion, ϵ is the target photon energy and m_p is the proton rest mass energy. Consider the collision to be head-on.

(b) Obtain the value of the threshold total energy of the proton for an interaction on a cosmic microwave background photon.

Hint: Similar to that of Question 1